

JOHAN A. VIJAYAN

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SUMMARY

Curious and resourceful Computer Science graduate with hands-on experience in full-stack development, backend development, machine learning, blockchain, and cloud-based systems. Skilled in designing and deploying scalable APIs, web applications, and ML-powered solutions. Thrive on building reliable, user-friendly software, optimizing performance, and solving complex technical challenges. Passionate about continuous learning and contributing to impactful technology.

EDUCATION

Illinois Institute of Technology

M.S. in Computer Science

Chicago, IL

August 2023 - May 2025

- **GPA:** 3.2
- **Relevant Courses:** Software Engineering, Advanced Machine Learning, Statistical Learning, Software Project Management, Computer Networks, Big Data Technologies, Information Security, Data Analysis and Engineering

Awards & Honors:

- Best Student in Computer Networks - 1, IIT Chicago (Top in 10 years)

University of Mumbai

B.E. in Computer Engineering

Mumbai, MH

August 2019 - May 2023

- **GPA:** 3.7
- **Relevant Courses:** Data Structures, Analysis of Algorithms, Object-Oriented Programming (OOP), Artificial Intelligence, Computer Architecture, Distributed Computing, Cybersecurity, Cryptography, Big Data Analytics

PROFESSIONAL EXPERIENCE

EVU Ventures - Cheaper

Software Developer

New York, NY

March 2025 – Present

- Built and maintained scalable RESTful APIs using Python, Flask, and Django to integrate modular cloud pricing scrapers.
- Designed abstract base classes (ABCs) to enforce consistent scraper architecture, enhancing code reuse and extensibility.
- Led backend development enhancements and containerized backend services with Docker and deployed them using Gunicorn and WSGI, ensuring production-ready, fault-tolerant infrastructure with seamless scalability.
- Mentored 10+ contributors on coding standards, best practices, and efficient deployment workflows.
- Refactored scraper logic, improving modularity and fault tolerance, enabling integration of 3+ future cloud providers.
- Developed unit and integration tests with CI pipelines to ensure robustness, reliability, and smooth deployments.

Tech Stack: Python, Flask, Django, Docker, Gunicorn, WSGI, CI/CD (GitHub Actions), Git

PROJECTS & OUTSIDE EXPERIENCE

ArcPay – Real-Time Biometric Payment Authentication System

MLH- Northwestern University

- Trained a custom CNN on 10,000+ palm images using triplet loss.
- Implemented liveness detection and micro-texture analysis with OpenCV to prevent spoofing.
- Developed TensorFlow/Keras embedding model optimized for low-latency, real-time authentication (<500ms)

Tech Stack: Python, TensorFlow, Keras, OpenCV, JavaScript, HTML, MongoDB Atlas.

FundsFly – MLH hackathon winner 2024

MLH - Northwestern University

- Built a Starknet L2 blockchain for real-time, offline-capable transactions using zero-knowledge proofs (ZKPs).
- Engineered a secure protocol for offline transaction validation, reducing latency by 40% versus traditional methods.
- Demonstrated decentralized scalability with 1K+ TPS in simulated environments.

Tech Stack: Python, Kotlin, Solidity, Starknet, ZKPs

FitBot – AI-Based Fitness and Nutrition Platform

Scarlet Hacks - Illinois Institute of Technology

- Built a JavaScript & Node.js platform integrating ChatGPT API to automate personalized diet and fitness plans. Tailored workout programs dynamically for diverse users with progress tracking.
- Integrated real-time video consultations via WebRTC and secure JWT-authenticated scheduling system.

Tech Stack: Node.js, JavaScript, ChatGPT OpenAI-API, WebRTC, JWT, MongoDB

Deep Learning – Software Defect Prediction

Remote

Personal Project

- Built a CNN-based deep learning model to predict software defects, reducing testing efforts by 30%.
- Trained on a subset of the Promise Dataset (10K+ samples) using TensorFlow and scikit-learn, achieving 92% precision.
- Optimized model via hyperparameter tuning and cross-validation, reducing validation costs by 25%.

Tech Stack: Python, TensorFlow, scikit-learn, Pandas, NumPy

Machine Learning – Attendance System with Facial Recognition

Remote

Personal Project

- Developed a scalable CNN-based facial recognition attendance system with 95% real-time accuracy.
- Designed a PyQt interface for user management; integrated cloud-based agent communication for distributed use.
- Enhanced security with liveness detection and optimized latency to <500ms per recognition.

Tech Stack: Python, TensorFlow, PyQt, OpenCV, Cloud services

SKILLS AND INTERESTS

Languages: Python, JavaScript, TypeScript, SQL, Bash, HTML/CSS, Dart

OS: Linux, Windows, Macintosh

Frameworks & Libraries: Flask, Django, Node.js, React.js, TensorFlow, PyTorch, scikit-learn

LLM Frameworks and ML: LangChain, AutoGPT, OpenAI API, CNN, ANN, prompt engineering

Blockchain: Starknet, Ethereum, Layer-2 solutions

Cloud & DevOps: AWS (S3, EC2), Google Cloud, Docker, CI/CD (GitHub Actions, Gunicorn), Logging & Monitoring

Tools: Cursor, VS Code, Jupyter, Anaconda, Colab, MS Project, Jira, Redis, Elasticsearch

Spoken Languages: English (Fluent)

HACKATHONS AND AWARDS

- **Wildhacks 2024 Winner** – Northwestern University | Built Starknet-based blockchain solution achieving 1K+ TPS
- **Best Financial Hack Award** (CapitalOne/Starknet) | Recognized for innovative L2 transaction protocol
- **HackOverflow Organizer** – Mumbai | Led 200+ participants in 24-hour hackathon
- **COMPUTER NETWORKS** - Illinois Institute of Technology | Awarded for best student in last 10 years.